

Ethan Cratchley

778-837-0285 | ethancratchley@gmail.com | linkedin.com/in/ethan-cratchley | github.com/EthanCratchley | ethancratchley.com

Education

Simon Fraser University (SFU)

Sep 2022 – Apr 2027

Bachelor of Computer Science and Business Administration (Joint Major) - Concentration in Artificial Intelligence

- **Coursework:** Data Structures & Algorithms, Artificial Intelligence, Database Engineering, Systems Programming, Software Engineering, Calculus, Linear Algebra, Discrete Mathematics, Statistics, Product Management

National University of Singapore (NUS)

Aug 2025 – Dec 2025

Exchange Program

Experience

Software Engineer

Jun 2026 – Present

Zyrka

Remote

- Cut inventory-check latency 75%+ under concurrent load via async write offloading, event-queue batching, and location-scoped WebSocket routing.
- Shipped backend features for a vertical SaaS platform automating dealership lot management, working directly with the founding team to turn customer needs into production code.

Venture Scout

Apr 2026 – Present

Front Row Ventures

Vancouver, BC

- Sourced deals at startup and ecosystem events, took founder meetings, and presented evaluations in the investment committee alongside the partner team.
- Conducted due diligence and market research on early-stage startups, assessing market size, traction, and scalability to surface high-potential deals.
- Built an AI-powered sourcing engine to automate deal discovery and surface qualified early-stage startups.

Software Engineer

Dec 2024 – Present

Autonommi

Hybrid

- Deployed scalable AI APIs enabling real-time inference and integration across multiple product environments, ensuring high availability and low latency for user workloads.
- Improved model inference speed by +50% and reduced cloud compute and data transfer (ingress/egress) costs by ~25% through caching, batching, and optimized data pipelines.

Innovation Intern

Apr 2024 – Aug 2024

Wilson Beck Insurance

Vancouver, BC

- Implemented AI-driven automation workflows that enhanced internal operational efficiency and boosted LLM performance.
- Researched and streamlined internal procedures, increasing employee efficiency by ~15%.
- Developed and deployed Python based automation tools to optimize company operations and workflows.

Projects

Kaggle Competitions & Writeups | Python, PyTorch, Scikit-learn, Feature Engineering, Data Analysis

Feb 2025 - Present

- Authored high-quality Kaggle notebooks and writeups with 20K+ views, 5K+ downloads, and 50+ upvotes, demonstrating strong community engagement and reproducible research practices.
- Implemented and benchmarked a range of models such as CNNs, Transformers, XGBoost, SVMs, Linear Regression, and more using Python, PyTorch, TensorFlow, and Scikit-learn to solve real-world datasets and competition problems.

Custorix | Python, Flask, Gemma 3n, YOLOv11, Raspberry Pi, Ollama

Jun 2025 - Aug 2025

- Built a private, offline first, multimodal AI security system using Gemma 3n, YOLOv11, and Raspberry Pi to deliver real-time video/audio analysis and natural language alerts without cloud dependencies.
- Optimized model deployment on resource-constrained Raspberry Pi hardware, enabling 30fps local inference.

Skills, Honors, Certifications, and Organizations

Skills: System Design, REST APIs, Data Visualization, MLOps, Testing & Debugging (Unit, Integration), CI/CD

Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL, HTML, CSS, Bash, LaTeX

Frameworks & Libraries: PyTorch, TensorFlow, Langchain, Pinecone, React, Next, Node, Express, Flask, Scikit-learn, FastAPI

Tools: Git, AWS, Google Cloud, MongoDB, Redis, Docker, Kubernetes, Figma, PowerBI, Supabase, PostgreSQL, Terraform

Competitions & Honors: Memorial International Study Abroad Award, WinterHacks Best Pitch, CrackedDevs Best Educational Project, BMO Marathon Completion, Gruesome Twosome Half Marathon 7th Place, CFAC Portfolio Management, PicoCTF